

Mycobacterium leprae — Leprosy (Hansen Disease) Sketch



1. Armadillo Co. Jail

WHAT YOU SEE

Sign reading 'Armadillo Co. Jail'

WHAT IT ENCODES

The armadillo is the classic animal reservoir of *Mycobacterium leprae*; the bacillus prefers cool body temperatures. Transmission to humans is via prolonged contact and respiratory droplets, not casual contact. *M. leprae* cannot be cultured on artificial media.

BOARD TRAP

M. leprae is the ONLY mycobacterium that does NOT grow on artificial culture media — diagnosis is by skin biopsy/slit-skin smear, not culture; don't expect a positive culture.

2. Armadillo reservoir

WHAT YOU SEE

Armadillo standing near the jail

WHAT IT ENCODES

Armadillos are a natural zoonotic reservoir of leprosy, especially in the southern US; the organism thrives in their low core temperature. This mirrors the bacterium's preference for cooler human sites (skin, peripheral nerves, nose, earlobes).

BOARD TRAP

Cool-loving bacillus → it targets cooler tissues (skin, ulnar/peroneal nerves, nasal mucosa, earlobes), sparing warm core organs — that distribution is the exam clue.

3. Deputy sheriff badge

WHAT YOU SEE

'Deputy Sheriff' star badge

WHAT IT ENCODES

The acid-fast bacillus is identified with Fite or Ziehl-Neelsen stain on biopsy. *M. leprae* is an obligate intracellular organism that invades Schwann cells of peripheral nerves.

BOARD TRAP

It's acid-fast like *M. tuberculosis* but is intracellular in SCHWANN CELLS — nerve involvement (not lung) dominates the picture.

4. Lepromatous prisoner (low immunity)

WHAT YOU SEE

Man behind bars, weak/imprisoned

WHAT IT ENCODES

Lepromatous (multibacillary) leprosy reflects a weak Th2-skewed cell-mediated response: numerous bacilli, symmetric widespread skin lesions, leonine facies, saddle-nose, and a low/negative lepromin skin test. Treated with rifampin + dapsone + clofazimine.

BOARD TRAP

LEPROMATOUS = poor Th1/CMI, HIGH bacterial load, lepromin-NEGATIVE — opposite of tuberculoid; don't swap their immune profiles.

5. Tuberculoid figure (strong immunity)

WHAT YOU SEE

Stronger, more contained figure

WHAT IT ENCODES

Tuberculoid (paucibacillary) leprosy reflects a robust Th1/cell-mediated response: few bacilli, well-demarcated hypopigmented anesthetic plaques, granulomas, and a POSITIVE lepromin test. Treated with rifampin + dapsone.

BOARD TRAP

TUBERCULOID = strong Th1/CMI, FEW bacilli, lepromin-POSITIVE, localized lesions — the mirror image of lepromatous.

6. Raised hands — anesthetic skin

WHAT YOU SEE

Man with hands raised, surrender

WHAT IT ENCODES

Hallmark of leprosy is hypoesthetic/anesthetic skin lesions from peripheral nerve invasion; loss of sensation precedes secondary trauma and deformity.

BOARD TRAP

A HYPOPIGMENTED patch with LOSS OF SENSATION = leprosy; a numb skin lesion is the key discriminator from vitiligo or tinea (which are NOT anesthetic).

7. Nerve thickening / claw hand

WHAT YOU SEE

Deformed hand/limb in a struggle

WHAT IT ENCODES

Schwann-cell invasion causes palpable thickened peripheral nerves (ulnar, common peroneal, great auricular) and motor deformities like claw hand, foot drop, and resorption of digits from repeated unnoticed injury.

BOARD TRAP

Palpable, THICKENED peripheral nerves with a numb skin lesion are nearly diagnostic of leprosy — few other diseases enlarge nerves this way.

8. Cowboy on horseback (type 1 reaction)

WHAT YOU SEE

Armed rider charging on a horse

WHAT IT ENCODES

Leprosy reactions: type 1 (reversal) reaction is a sudden cell-mediated flare causing inflamed skin lesions and acute neuritis — a medical emergency treated with corticosteroids to prevent permanent nerve damage.

BOARD TRAP

Acute neuritis/worsening lesions during therapy = TYPE 1 reversal reaction → give STEROIDS, do NOT stop antibiotics.

9. Drawn gun (erythema nodosum leprosum)

WHAT YOU SEE

Raised pistol about to fire

WHAT IT ENCODES

Type 2 reaction = erythema nodosum leprosum (ENL), an immune-complex (Th2) reaction with tender skin nodules, fever, and systemic illness, mainly in lepromatous disease; treated with corticosteroids or thalidomide.

BOARD TRAP

ENL (type 2) responds to THALIDOMIDE — a classic association — but thalidomide is teratogenic and contraindicated in pregnancy.

10. Window shooter (dapsone)

WHAT YOU SEE

Man firing through the jail window

WHAT IT ENCODES

Dapsone is a backbone of multidrug therapy (inhibits folate synthesis). It can cause hemolysis, especially in G6PD deficiency, and methemoglobinemia.

BOARD TRAP

Screen for G6PD deficiency before DAPSONE — it precipitates hemolytic anemia in G6PD-deficient patients.

11. Clofazimine red-brown skin

WHAT YOU SEE

Reddish-brown earth tones / clothing

WHAT IT ENCODES

Clofazimine is added for multibacillary disease and has anti-inflammatory effects; its hallmark side effect is reddish-brown skin discoloration that is reversible.

BOARD TRAP

Reddish-brown skin pigmentation in a leprosy patient = CLOFAZIMINE effect, reversible — not disease progression.

12. Rifampin (rifa = red)

WHAT YOU SEE

Red-orange brick wall / tones

WHAT IT ENCODES

Rifampin is the most bactericidal antileprosy drug and the core of all regimens; it turns body fluids red-orange and induces CYP450, lowering levels of many co-administered drugs.

BOARD TRAP

Rifampin's red-orange secretions are harmless, but its CYP450 INDUCTION can fail oral contraceptives and lower other drug levels — counsel accordingly.

13. Saddle-nose collapse

WHAT YOU SEE

Facial/nasal area of central figure

WHAT IT ENCODES

Advanced lepromatous leprosy destroys nasal cartilage producing saddle-nose deformity and leonine facies from diffuse dermal infiltration; the nose's cool mucosa is a favored site.

BOARD TRAP

Saddle-nose has a differential (congenital syphilis, GPA/Wegener's) — in leprosy it's accompanied by anesthetic skin and nerve thickening, the distinguishing combo.

14. Multidrug therapy duration

WHAT YOU SEE

Group standoff implying a regimen team

WHAT IT ENCODES

WHO multidrug therapy prevents resistance: paucibacillary = rifampin + dapsone (6 months); multibacillary = rifampin + dapsone + clofazimine (12 months). Combination therapy is mandatory.

BOARD TRAP

Never treat leprosy with monotherapy — single-drug therapy breeds resistance; multibacillary needs the THREE-drug regimen for 12 months.
